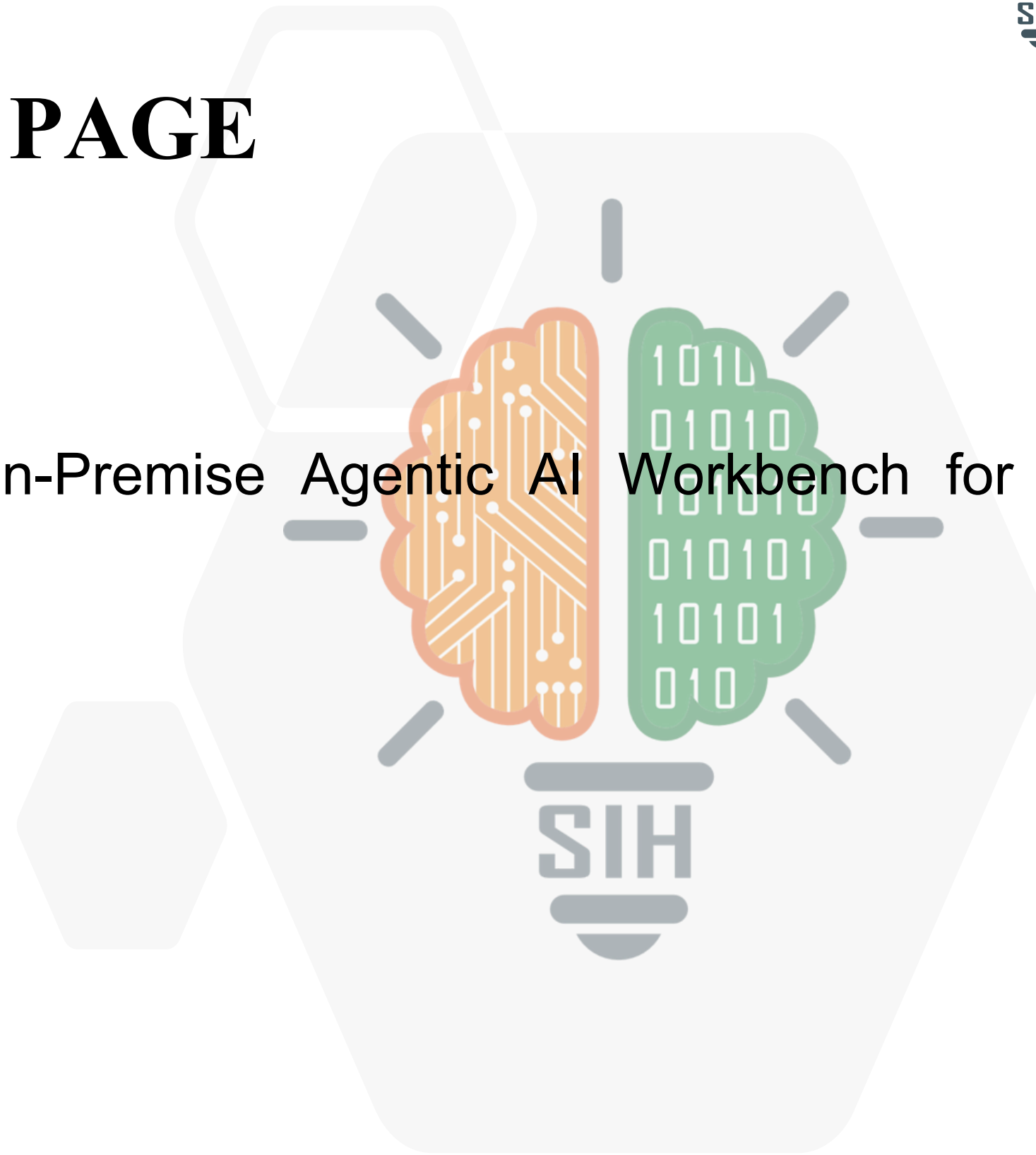


TITLE PAGE

- **Problem Statement ID** – 26117
- **Problem Statement Title-** Sovereign On-Premise Agentic AI Workbench for Confidential Industrial Intelligence
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name** – NEXORA



Problem:

Industrial organizations work with **sensitive engineering, financial, inspection, and internal data** that cannot be shared with **public AI tools**. This leaves teams dependent on **manual work**, slowing down **analysis** and reducing **productivity**.

Proposed Solution :

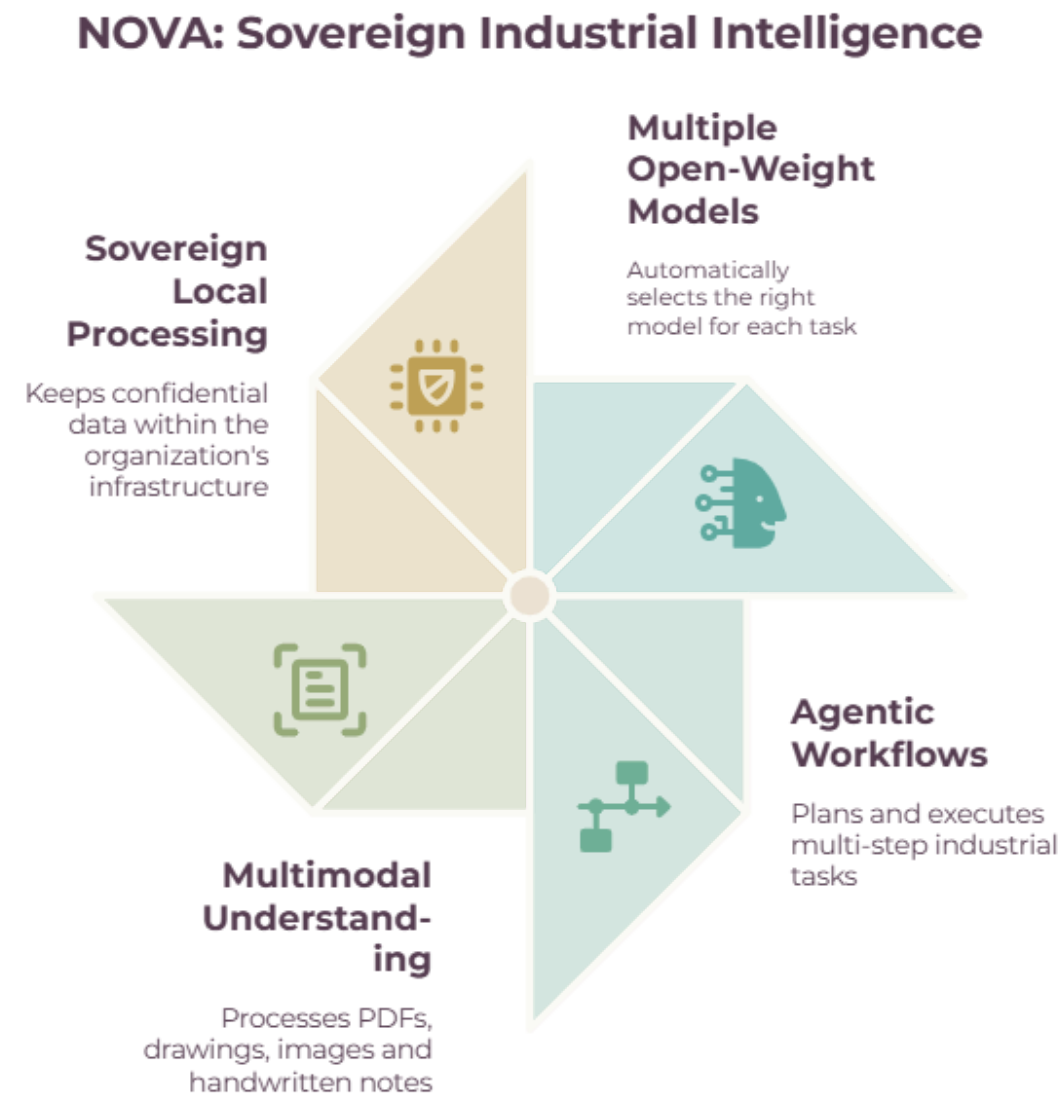
- A **self-hosted AI workbench** runs open-weight models locally and automatically selects the **most suitable model for each task**, without sending confidential data outside the organization.
- An **agentic workflow** plans and performs multi-step tasks using **local file tools, internal knowledge search, sandboxed code execution, and iterative reasoning**.
- A **multimodal AI system** understands **scanned PDFs, handwritten notes, engineering drawings, images, and documents**, and generates **practical work outputs** such as reports, approval notes, calculations, presentations, and code.

Our Idea :

NOVA is a **self-hosted sovereign AI workbench** that runs **open-weight multimodal models** on the organization's own infrastructure. It enables **secure local AI** for confidential industrial work, with **automatic model selection** and **agentic workflows** that can understand **documents**, use **local tools**, and generate **practical outputs**.

Innovation/Uniqueness:

- **Model-agnostic AI** supports multiple **open-weight models** with automatic task-based model selection.
- **Agentic workflow** plans and executes **multi-step industrial tasks** using local tools and enterprise knowledge.
- **Multimodal AI** understands **scanned documents, drawings, images, and handwritten notes** for industrial use.
- **Sovereign deployment** keeps sensitive data **on-premise and air-gapped** with no external data flow.

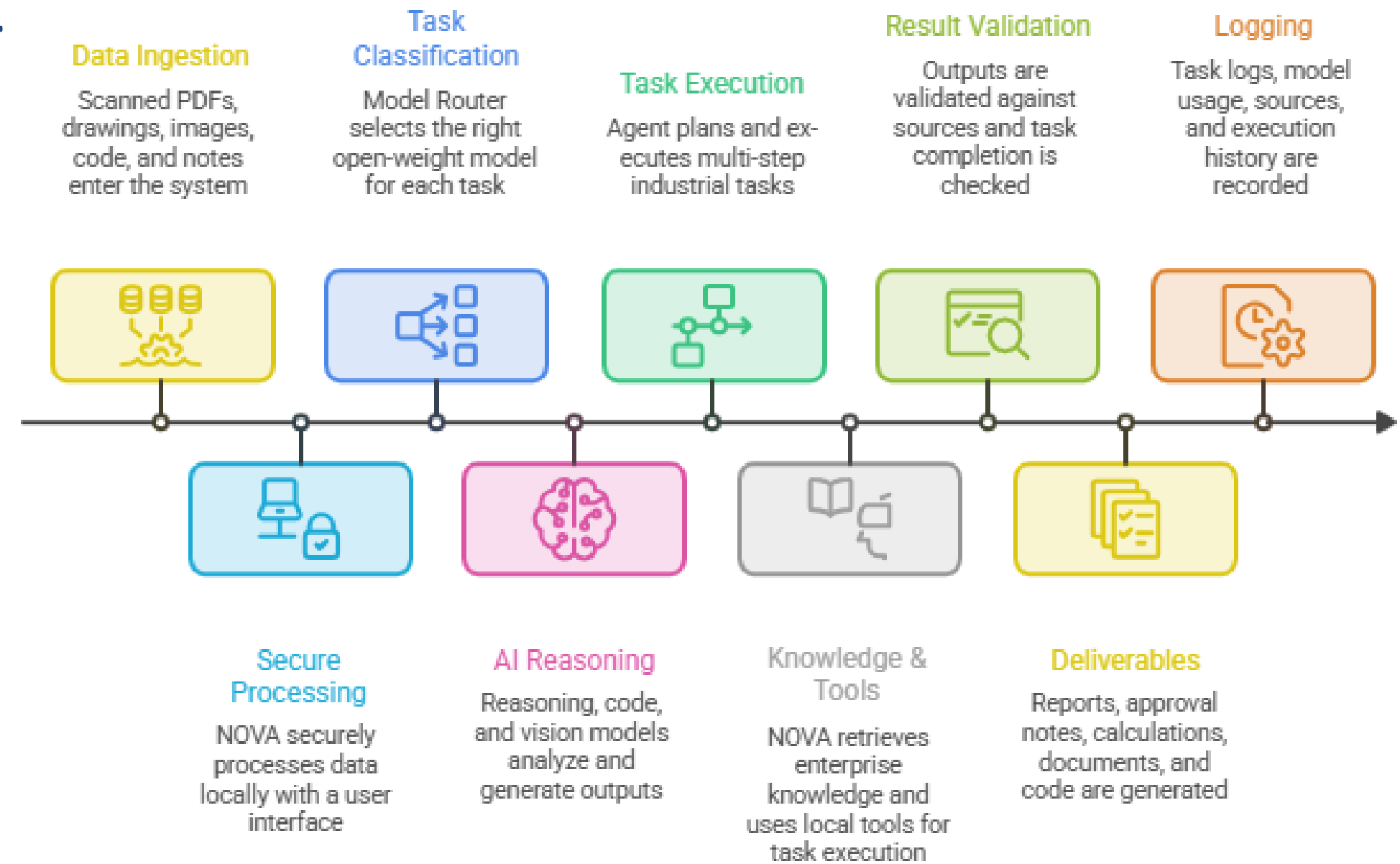


TECHNICAL APPROACH

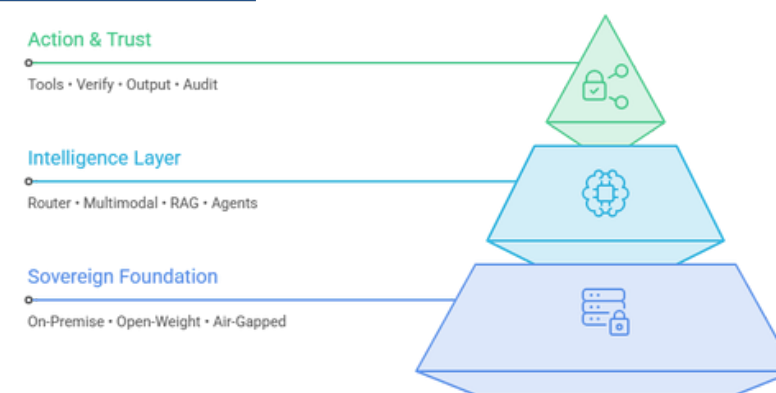
Hardware & Software

- **GPU Workstation/Server:** Runs local AI models and on-premise inference.
- **Python / FastAPI:** Handles AI orchestration, APIs, and backend services.
- **React:** Provides the interactive AI workbench.
- **Ollama + LLMs:** Runs open-weight models locally.
- **LangGraph:** Manages multi-step agent workflows.
- **ChromaDB:** Provides local semantic search over enterprise knowledge.
- **OCR / Vision Models:** Process scanned PDFs, drawings, images, and handwritten notes.
- **Local Tools / Sandbox:** Enables file operations, code execution, and document processing.

FLOW CHART



3 LAYER APPORACH



GitHub link:

YouTube video : [Link](#)

Feasibility

- **Local deployment:** Runs on a single workstation/GPU server using open-weight models.
- **Modular architecture:** Supports adding new models and tools without redesigning the system.
- **Local processing:** RAG and AI inference run within the organization's infrastructure.
- **Controlled execution:** Sandboxed tools and validation help ensure safe agent actions.
- **Scalable design:** Supports mid-range GPU deployment with scope for future scaling.

Challenges

- **Data privacy:** Confidential industrial data must remain secure and local.
- **Model accuracy:** Open-weight models may vary in task performance.
- **Multimodal processing:** Documents, drawings, and images require reliable AI understanding.
- **Agent reliability:** Multi-step agents need validation and controlled execution.

Commercial feasibility

- **Deployment Readiness:** Suitable for on-premise deployment within industrial organizations.
- **Cost Efficiency:** Uses open-weight models and local infrastructure, reducing dependence on paid cloud AI services.
- **Enterprise Adoption:** Supports confidential workflows across refineries, PSUs, defence-linked units, and government offices.
- **Scalability:** Modular design allows new models and tools to be added as requirements evolve.

Strategy:

- **Model routing:** Automatically selects the best open-weight model for each task.
- **Agent orchestration:** Uses multi-step planning and execution for industrial workflows.
- **Knowledge retrieval:** Uses local RAG to access relevant enterprise documents.
- **Controlled execution:** Uses local tools, sandboxing, and validation for safe task completion.

Direct Impact on Target Users

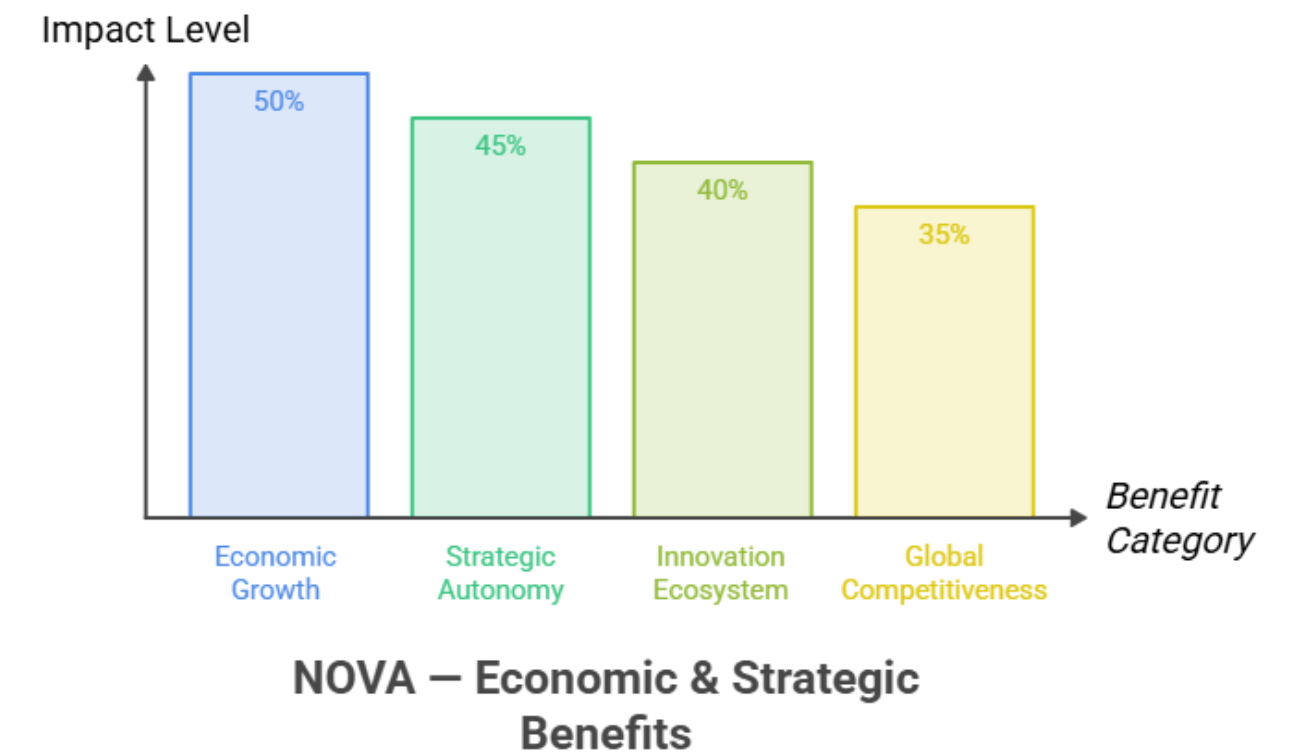
- **Refineries & PSUs:** Enables secure AI assistance for confidential engineering and operational work.
- **Defence-linked Industries:** Supports protected analysis of sensitive documents, designs, and internal data.
- **Government Offices:** Reduces manual effort in reports, approvals, analysis, and document processing
- **Industrial Teams:** Provides local AI workflows for faster analysis while keeping confidential data within the organization.

Strategic Impact

- **Strengthens data sovereignty** by enabling **on-premise, air-gapped** AI for confidential industrial work.
- Reduces dependence on **external cloud AI services** while supporting **open-weight multimodal models**.
- Enables **secure agentic automation** across engineering, inspection, documentation, coding, and other industrial workflows.
- **Builds an extensible AI foundation** where new models, tools, and capabilities can be integrated without redesigning the system.

Economic & Strategic Benefits

- **Reduces cloud AI dependency** by using **open-weight models and local infrastructure** for industrial workloads.
- **Improves operational efficiency** by automating **analysis, documentation, coding, and repetitive knowledge work**.
- **Strengthens India's AI sovereignty** through **self-hosted, secure industrial intelligence**.
- **Supports scalable adoption** across **refineries, PSUs, defence-linked industries, and government offices**.



Gap & Problem Identification

- **Growing enterprise AI adoption** is increasing the need for **secure handling of sensitive organizational data**.
- **Public AI tools create data-security concerns**, while organizations need **controlled, on-premise AI** for confidential industrial workflows.

Literature Survey & Competitive Analysis

- **Existing AI assistants** rely largely on cloud infrastructure, creating **privacy and confidentiality concerns** for sensitive industrial work.
- **NOVA differentiates through on-premise agentic AI**, combining **open-weight models, multimodal processing, local knowledge, and tool-based execution** in one workbench.

Technology Benchmarking

- **Compare cloud-based AI assistants** with NOVA on **data privacy, deployment control, model flexibility, and local processing**.
- Highlight NOVA's integration of **open-weight models, agentic workflows, multimodal AI, and local processing.knowledge/tools** in a single on-premise workbench.

Economic & Strategic Landscape

- Growing demand for **secure enterprise AI** highlights the need for **on-premise solutions** for sensitive industrial operations.
- Increasing adoption of **open-weight and agentic AI** creates opportunities for **sovereign industrial AI platforms** across refineries, PSUs, defence-linked units, and government offices.

Field Tests & Simulation Results

- **Prototype Testing:** Demonstrate NOVA on a **local GPU workstation** across document, coding, and multimodal industrial tasks.
- **Workflow Validation:** Evaluate **model selection, agent execution, and output verification** using representative confidential-work scenarios.

Policy & Ecosystem Analysis

- Aligns with India's AI sovereignty goals by enabling **secure on-premise AI** for sensitive industrial and government workflows.

